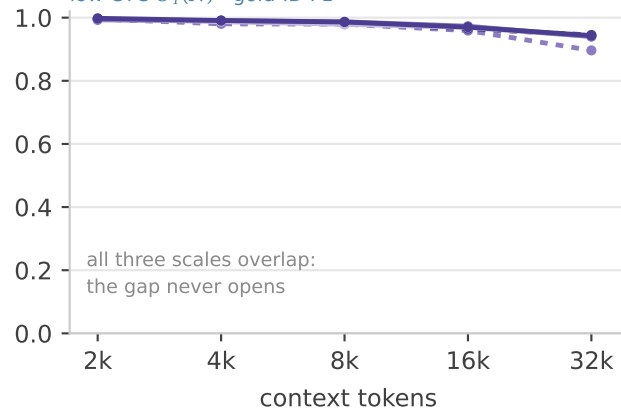


Scaling 0.8B $\rightarrow$ 4B does not close the chunked-attention gap on high-CTC tasks

Qwen3.5 at three parameter counts, identical training shards and identical evaluation ladders; solid = full attention, dotted = chunked

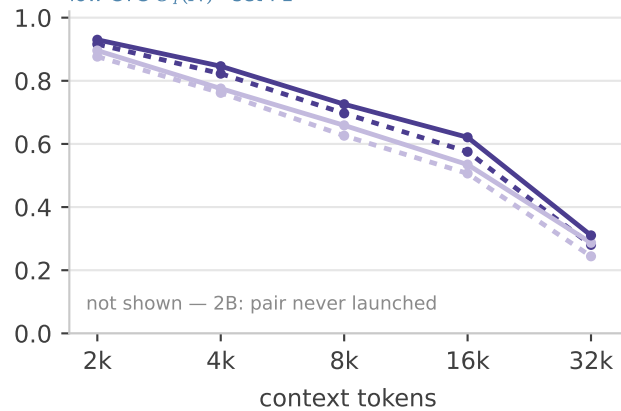
(a) HotpotQA

low CTC $O_T(N)$ · gold-ID F1



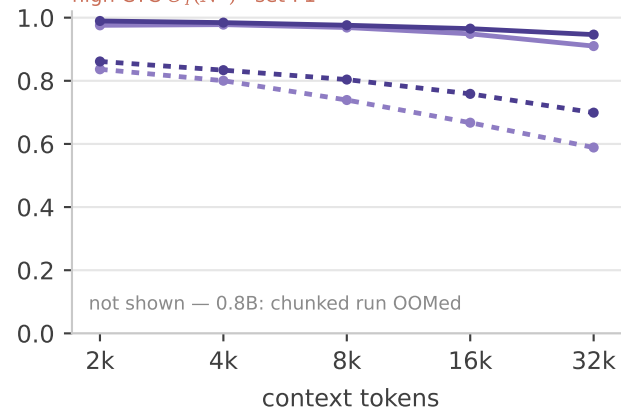
(b) BEIR FiQA

low CTC $O_T(N)$ · set-F1



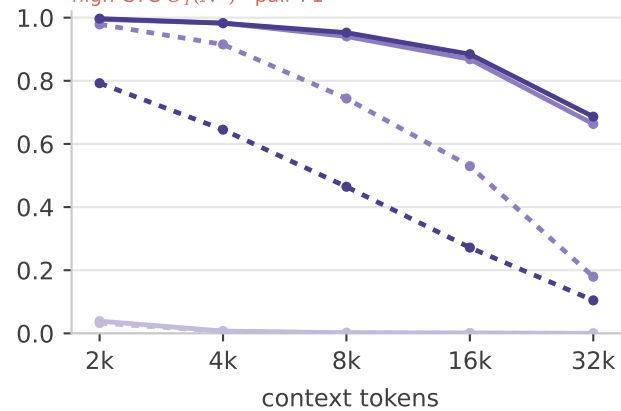
(c) contradiction

high CTC $O_T(N^2)$ · set-F1



(d) qdmach (NQ)

high CTC $O_T(N^2)$ · pair-F1



Qwen3.5-0.8B Qwen3.5-2B Qwen3.5-4B full attention chunked attention